BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023 • Paper Code: MRD-9th 110525 (A)

Retyped faithfully from the original board paper — same questions, no changes.

Section – A

Time Allowed: 20 Minutes | **Maximum Marks:** 15 | **Paper Code:** MRD-9th 110525 (A), VERSION: B

Note: There are three sections in this paper i.e A, B & C. Instructions: Please attempt Section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, over writing, erasing, or multiple circles shading.

Q1. Choose the correct option i.e A, B, C or D.

Q1. $(m - 3n)^3 =$

(A) $(m^3+9n^3+9mn)/(m+3n)$
(B) $(m^3+9n^3+9mn)/(m-3n)$
(C) $(m^3-9n^3-9mn)/(m-3n)$
(D) $(m^3-9n^3+9mn)/(m+3n)$

Q2. The factorization of $u^2 - 6uv + 9v^2$ is

(A) $(u-3v)(u-3v)$
(B) $(u+3v)(u+3v)$
(C) $u(u-6v)+9v^2$
(D) $u^2-3v(2u-3v)$

Q3. If C and D are any two polynomials and their LCM is L, then their HCF =

(A) L/(CxD)
(B) CxD/L
(C) C/L
(D) D/L

Q4. The solution of linear equation $2x + 5 = x - 3$ is:

(A) -8
(B) -8
(C) 1
(D) -1

Q5. If A = {5,4} and B = {2,3}, then which of the following is ordered pair of A x B)?

(A) {(2,5),(3,4)}
(B) {(5,4),(2,5)}
(C) {(5,2)}
(D) {(2,4)}

Q6. Which of the following elements represent one of the columns of matrix |11 9 7; 6 −5 4; 5 6 3|?

(A) 11, 9, 7
(B) 5, 6, 3
(C) 11, 5, 3
(D) 7, 4, 3

Q7. Which of the following two matrices are equal: S = |7 9; 1 8|, T = |3+5 3+6; 1 8|, U = |3+6 4+6; 1 8|, V = |3+4 3+6; 1 8|

(A) S and T
(B) T and U
(C) S and V
(D) U and V

Q8. Which of the following is a diagonal matrix?

(A) |1 2; 4 4|
(B) |1 0; 0 4|
(C) |0 2; 4 4|
(D) |1 2; 0 4|

Q9. If A = |11 2; −3 −4| and B = |1 7; 5 −8| then A−B is equal to:

(A) |12 9; 8 12|
(B) |10 −5; −8 4|
(C) |11 14; 15 28|
(D) |10 5; 2 −4|

Q10. The rational number 2/15 can be e::pressed as:

(A) 0.333...
(B) 1.333...
(C) 0.1333...
(D) 0.0333

Q11. $^{3}/_{z_5}$ can be expressed in radical form as:

(A) $\sqrt[3]{z_2}$
(B) $\sqrt[5]{z^3}$
(C) $\sqrt[5]{z}$
(D) $\sqrt[3]{z}$

Q12. If z = 11 + 9i the conjugate of z is:

(A) −11 − 9i
(B) −11 + 9i
(C) 11 − 9i
(D) 11 + 9i

Q13. 3.21 x 10^4 can be written in standard form as:

(A) 3210
(B) 32100
(C) 321000
(D) 3210000

Q14. Which of the following is the base of Common logarithm?

(A) e
(B) G
(C) 10
(D) e

Q15. Which of the following is a polynomial?

(A) $6x^4 + 3x^3 + 3$
(B) $6x^{-4} + 3x^{-3} + 4$
(C) $6x^4 + 3x^{-3} + 4$
(D) $6x^{-4} + 3x^{-3} + 3$

Section – B

Time Allowed: 2.40 Hours	**Maximum Marks:** 36

Q2. Attempt any nine (9) parts:-

(i) If E = |7 2; 8 4|, F = |15 −7; 3 −2| determine whether EF = FE, or not?

(ii) Simplify: (a) $(9l^3m^2)^5$ (b) $(-2x^2y^3)^3$

(iii) Find the value of 'l': $\log_4 9 + \log_4 7 - \log_4 3 = \log_4 l$

(iv) Evaluate the following when y = − 4: $(5y^2\ 3y-2)/(y-2)$

(v) Find the value of st if s + t = 12, s − t = 4.

(vi) Without performing division, find the remainder when $7h^3-6h^2+5h-3$ is divided by h + 2.

(vii) Simplify: $8/(z^2 - 4) \div 8/(z + 2)$

(viii) Solve : $\sqrt{(7x + 9)} - 2 = 8$

(ix) Solve and plot on number line: $(y-1)/4 \leq (1-x)/2$, $x \in Z$

(x) Prove that the points: P(1,3), Q(2,4) and R(3,3) are collinear.

(xi) Prove that: If two angles of a triangle are congruent, then the sides opposite to those angles are congruent.

(xii) Prove that : Any point on the right bisector of a line segment is equidistant from end points of the segment.

Section – C

Maximum Marks: 24

> Note: Attempt any three (3) questions:-

Q3. Prove that: If two opposite sides of a quadrilateral are congruent, then it is a parallelogram

Q4. Prove that: From a point outside a line, the perpendicular is the shortest distance from the point to the line.

Q5. Prove that: If a line segment intersects the two sides of a triangle in the same ratio then it is parallel to the third side.

Q6. Prove that: In a right-angled triangle, the square of the length of hypotenuse is equal to the sum of the squares of the lengths of the other two sides.

Q7. Prove that: Parallelograms on equal bases and having the same altitude are equal in area.

Q8. Construct ΔPQR, for: $mQR = 6.5Cm$, $m\angle P = 30°$ and $m\angle R = 60°$